Exponential Functions & Growth Models

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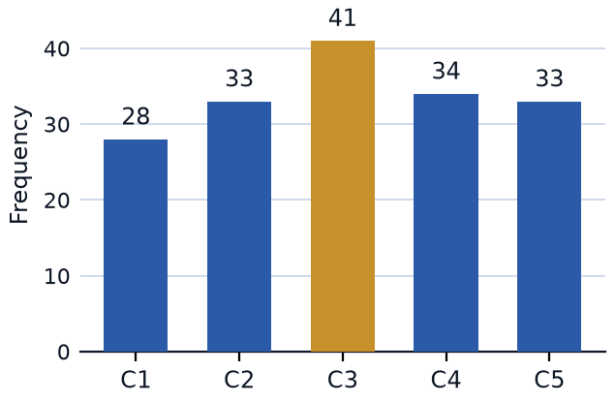
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Exponential growth and decay basics

Classify each function as exponential growth or decay, and state the growth/decay factor: $f(x) = 5(1.2)^x$ and $g(x) = 5(0.8)^x$.

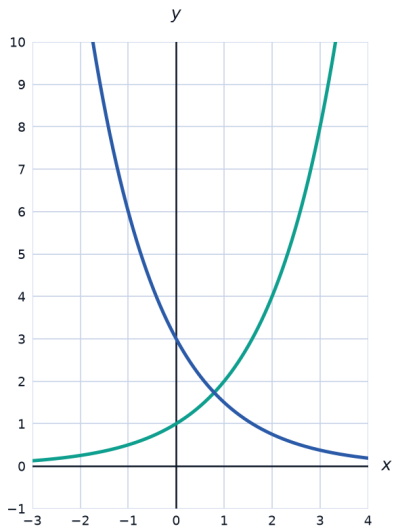
2 [Foundational] • Target: Exponential growth and decay basics

A population of bacteria is modeled by $P(t) = 1000(2)^{t/3}$, where t is in hours. Interpret the meaning of the exponent's structure, and find the population after 9 hours.



3 [Foundational] • Target: Visualizing exponential growth

The graph shows $f(x) = 2^x$ and $g(x) = 3(0.5)^x$, one growing and one decaying.



a Identify which curve represents growth and which represents decay.

b State the horizontal asymptote shared by both curves.

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4 [Foundational] • Target: Finding an exponential model from data

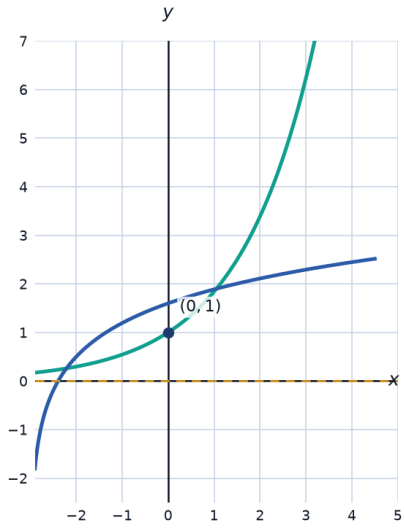
An investment grows from 2000 dollars to 2420 dollars in 2 years under exponential growth $A(t) = A_0(1 + r)^t$. Find the annual growth rate r.

5 [Foundational] • Target: Finding an exponential model from data

A radioactive sample decays from 80 g to 45 g over 6 years. Find the annual decay factor b in the model $A(t) = 80\,b^t$, to four decimal places.

6 [Foundational] • Target: Continuous growth and the number e

A quantity grows continuously according to $A(t) = 500e^{0.03t}$. Find the amount after 20 years, to the nearest whole number.



7 [Foundational] • Target: Continuous growth and the number e

Explain the difference between the discrete growth model $A(t) = A_0(1 + r)^t$ and the continuous growth model $A(t) = A_0e^{kt}$, and state how r and k are related when both models describe the same annual growth rate r.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Comparing growth rates: exponential vs. polynomial

Explain why, for large enough x, any exponential growth function $f(x) = b^x$ (with $b > 1$) will eventually exceed any polynomial function $g(x) = x^n$, no matter how large the exponent n is. Illustrate with $f(x) = 1.01^x$ vs. $g(x) = x^{10}$.

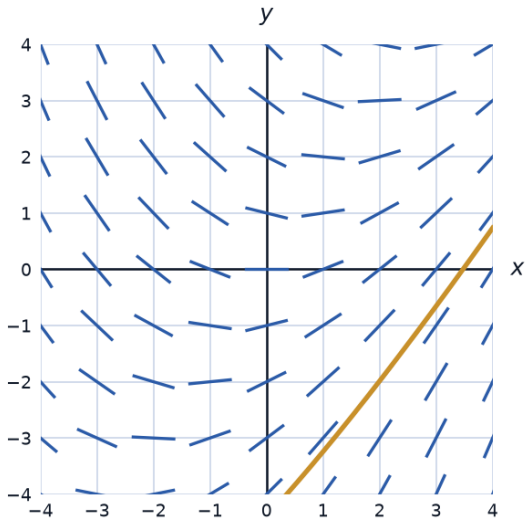
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9 [Intermediate] • Target: Applications: half-life and doubling time

A substance has a half-life of 8 days. Write its decay model $A(t) = A_0\left(\frac{1}{2}\right)^{t/8}$, then find what fraction of the original amount remains after 20 days.

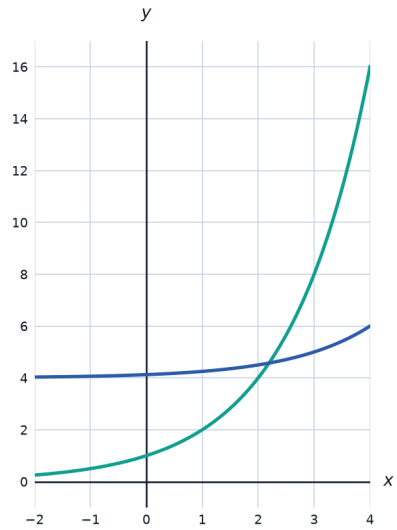
10 [Intermediate] • Target: Applications: half-life and doubling time

A city's population doubles every 15 years and currently stands at 40{,}000. Write an exponential model $P(t)$ and predict the population after 45 years.



11 [Intermediate] • Target: Transformations of exponential functions

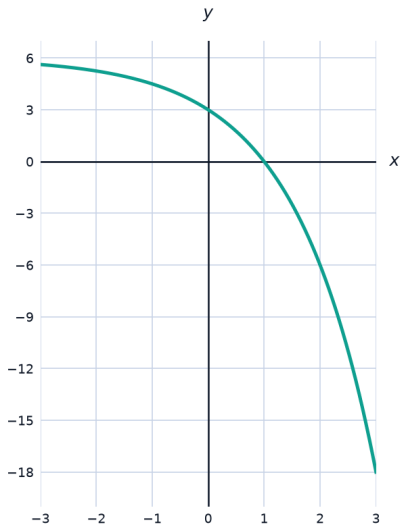
Describe how the graph of $g(x) = 2^{x-3} + 4$ is transformed from the parent function $f(x) = 2^x$, and state the new horizontal asymptote.



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12 [Intermediate] • Target: Transformations of exponential functions

Find the horizontal asymptote and y-intercept of $h(x) = -3(2)^x + 6$.



13 [Intermediate] • Target: Compound interest and financial models

A savings account earns 4% annual interest compounded quarterly, modeled by $A(t) = A_0\left(1 + \frac{0.04}{4}\right)^{4t}$. If 5000 dollars is deposited, find the balance after 3 years, to the nearest dollar.

14 [Intermediate] • Target: Compound interest and financial models

Explain, without computing exact values, why compounding more frequently (e.g. daily instead of quarterly) at the same nominal annual rate produces a slightly larger balance after the same number of years.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

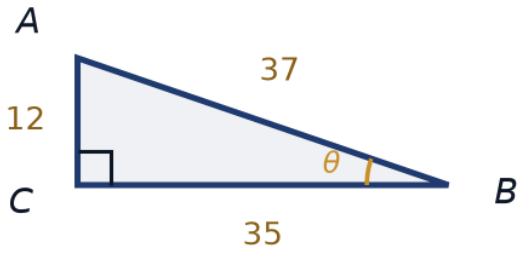
15 [Challenging] • Target: Transformations of exponential functions • Cross-Lesson: 4.03 Parametric Equations & Modeling

A radioactive substance follows the decay model $M(t) = M_0 e^{-kt}$. If 20% of the sample decays in 15 days, find the exact decay constant k and the substance's half-life in days to the nearest tenth.

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16 [Challenging] • Target: Compound interest and financial models • Cross-Lesson: 1.01 Polynomial End Behavior & Zeros

A Ferris wheel has a diameter of 40 meters and its lowest point is 4 meters above the ground. Passengers board at the lowest point at $t = 0$, and the wheel completes one full revolution every 8 minutes. Write a cosine model $h(t)$ for a passenger's height at time t, and find the times during the first revolution when $h(t) = 34$ meters.



17 [Challenging] • Target: Exponential growth and decay basics • Cross-Lesson: 1.02 Rational Functions & Asymptotes

The half-life of Carbon-14 is 5730 years. (a) Write $A(t)$. (b) How much of 200 g remains after 11 460 years? (c) When is only 10% left?

18 [Challenging] • Target: Visualizing exponential growth • Cross-Lesson: 1.03 Rates of Change & Concavity

[Unit 1 · Polynomial and Rational Functions · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

19 [Challenging] • Target: Finding an exponential model from data • Cross-Lesson: 2.02 Logarithmic Functions & Properties

[Unit 2 · Exponential and Logarithmic Functions · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

20 [Challenging] • Target: Continuous growth and the number e • Cross-Lesson: 3.01 Trigonometric Functions & the Unit Circle

Level 1 · FoundationsFind the exact value of $\tan\left(\frac{11\pi}{6}\right)$ and $\sec\left(\frac{11\pi}{6}\right)$ using reference angles and unit circle coordinates.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Exponential growth and decay basics

A population is modeled by $P(t) = 900(3)^{t/3}$, where t is in years. Find $P(6)$.

a. 2700

b. 5400

c. 8100

d. 24300

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22 [Foundational] • Target: Visualizing exponential growth

A substance has a half-life of 6 days. What fraction of the original amount remains after 18 days?

a. $\frac{1}{4}$

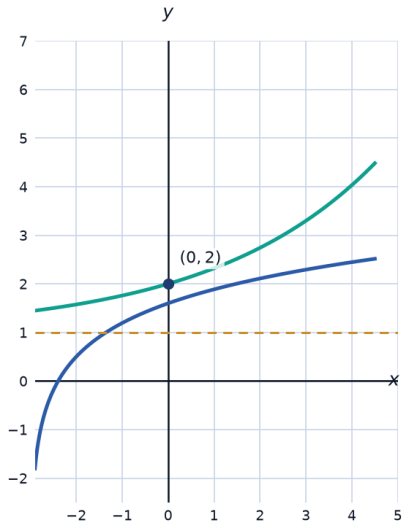
b. $\frac{1}{16}$

c. $\frac{3}{8}$

d. $\frac{1}{8}$

23 [Foundational] • Target: Finding an exponential model from data

An investment grows from 300 dollars to 507 dollars in 2 years under $A(t) = A_0(1 + r)^t$. Find the annual growth rate r.



a. $\frac{3}{10}$

b. $\frac{2}{5}$

c. $-\frac{3}{10}$

d. $\frac{13}{10}$

24 [Foundational] • Target: Continuous growth and the number e

A savings account earns interest modeled by $A(t) = A_0(1 + \frac{1}{20})^{nt}$, where $nt = 15$. If 2000 dollars is deposited, find the balance to the nearest dollar.

a. 4008

b. 4258

c. 2000

d. 4158

25 [Foundational] • Target: Comparing growth rates: exponential vs. polynomial

Which is an exponential decay?

a. $y = 3^x$

b. $y = 0.4^x$

c. $y = e^x$

d. $y = 10^x$

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26 [Foundational] • Target: Applications: half-life and doubling time

What is the half-life formula for exponential decay?

a. $A(t) = A_0(2)^{t/h}$

b. $A(t) = A_0e^{-t}$

c. $A(t) = A_0/2^t$

d. $A(t) = A_0\left(\frac{1}{2}\right)^{t/h}$

27 [Foundational] • Target: Transformations of exponential functions

If compound interest is $A = P(1 + \frac{r}{n})^{nt}$, which formula gives continuous compounding?

a. $A = Pe^{rt}$

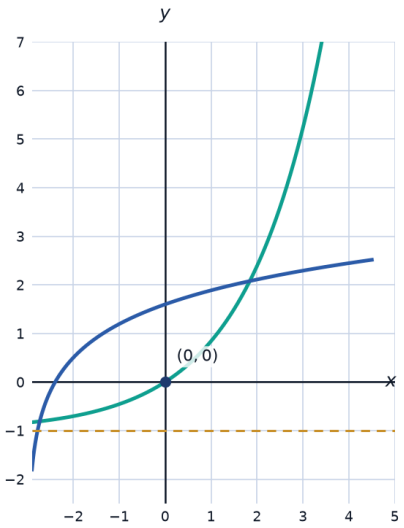
b. $A = P/e^{rt}$

c. $A = P(1 + r)^t$

d. $A = Prt$

28 [Foundational] • Target: Compound interest and financial models

The doubling time model is: $A = A_0 \cdot 2^{t/d}$. If $d = 5$, find A/A_0 at $t = 10$.



a. 8

b. 2

c. 4

d. 10

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Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 **[Intermediate]** • **Target: Exponential growth and decay basics**

For $f(x) = 2(0.8)^x$, the function represents:

a. exponential growth of 20% per unit x

b. no growth or decay (constant function)

c. linear change with rate 0.8 per unit x

d. exponential decay of 20% per unit x

30 **[Intermediate]** • **Target: Visualizing exponential growth**

Solve for x (round to 3 decimal places): $5^x = 15$.

a. 3.000

b. 2.183

c. 1.683

d. 1.183

31 **[Intermediate]** • **Target: Finding an exponential model from data**

By the Remainder Theorem, what is the remainder when $f(x) = 1x^3 + 5x^2 - 4x + 4$ is divided by $(x + 4)$?

a. 34

b. -36

c. 72

d. 36

32 **[Intermediate]** • **Target: Continuous growth and the number e**

A bacteria culture triples every 4 hours. Starting with 200 bacteria, the population after t hours is

a. $200 \cdot 3^{t/4}$

b. $200 \cdot 3^{4t}$

c. $200 + 3t$

d. $600 \cdot t$

33 **[Intermediate]** • **Target: Comparing growth rates: exponential vs. polynomial**

Remainder when $p(x) = x^3 - 2x + 1$ is divided by $(x - 2)$:

a. 3

b. 5

c. 7

d. 9

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34 [Intermediate] • Target: Applications: half-life and doubling time

$A = P(1 + r/n)^{nt}$. Find A (to the nearest cent) when x2000, r=0.04, n=2, t=2.

a. 2154.86

b. 2174.86

c. \2164.86

d. 2163.20

35 [Intermediate] • Target: Transformations of exponential functions

Consider the exponential growth model $P(t) = P_0e^{kt}$, where $P(t)$ is the population at time t and P_0 is the initial population. If the initial population is $P_0 = 500$ and after 10 years the population is $P(10) = 1200$, find the annual growth rate k. Round your answer to four decimal places.

a. 0.0588

b. 0.0875

c. 0.1200

d. 0.0434

36 [Intermediate] • Target: Compound interest and financial models

Consider the exponential growth model described by the function $P(t) = P_0e^{kt}$, where $P(t)$ is the population at time t, P_0 is the initial population, and k is the growth constant. If the initial population P_0 is 500 and the population after 4 years is 2000, determine the value of the growth constant k to three decimal places.

a. 0.347

b. 0.416

c. 0.301

d. 0.477

37 [Intermediate] • Target: Exponential growth and decay basics

For $A = A_0e^{-kt}$, find the half-life $t_{1/2}$ in terms of k.

a. $\frac{k}{\ln 2}$

b. $\frac{\ln 2}{k}$

c. $\frac{1}{k}$

d. e^k

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Visualizing exponential growth • Cross-Lesson: 1.03 Rates of Change & Concavity

[Transformation Trial 1] Identify the matrix representing a counterclockwise rotation by 360° in $\mathbb{R}^2$.

a. $(-1, 0; 0, 1)$

b. $(0, 1; 1, 0)$

c. $(1, 0; 0, 1)$

d. $(1, -1; 1, 1)$

39 [Challenging] • Target: Finding an exponential model from data • Cross-Lesson: 2.02 Logarithmic Functions & Properties

Pythagorean identity: $1 + \tan^2\theta =$

a. $\csc^2\theta$

b. $\sec^2\theta$

c. $\cot^2\theta$

d. $\cos^2\theta$

40 [Challenging] • Target: Continuous growth and the number e • Cross-Lesson: 3.01 Trigonometric Functions & the Unit Circle

Convert the rectangular point $(-1, 1)$ to polar form; give r and θ (in degrees, $0 \le \theta < 360$).

a. $r = 1.414,\ \theta = 225°$

b. $r = 2.414,\ \theta = 135°$

c. $r = 1.414,\ \theta = 315°$

d. $r = 1.414,\ \theta = 135°$

41 [Challenging] • Target: Comparing growth rates: exponential vs. polynomial • Cross-Lesson: 3.02 Polar Coordinates & Polar Graphs

Eliminate the parameter to find y as a function of x: $x = 1t - 2$, $y = -2t^2$.

a. $y = -1(x + 2)^2$

b. $y = -2(x + 2)^2$

c. $y = 2(x + 2)^2$

d. $y = -2(x - 2)^2$

42 [Challenging] • Target: Applications: half-life and doubling time • Cross-Lesson: 4.01 Inverse & Composite Functions

Write a sinusoidal function with amplitude 5, period 4π, midline $y = -3$, using $\cos$.

a. $f(x) = 5\cos(1/2x) - 3$

b. $f(x) = 5\cos(1/2x) - 2$

c. $f(x) = 6\cos(1/2x) - 3$

d. $f(x) = 5\cos(1x) - 3$

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43 [Challenging] • Target: Transformations of exponential functions • Cross-Lesson: 4.02 Piecewise-Defined & Implicit Functions

Identify the polar curve $r = 5 + 3\cos\theta$.

a. a cardioid

b. a circle

c. a dimpled limaçon

d. a rose with 3 petals

44 [Challenging] • Target: Compound interest and financial models • Cross-Lesson: 4.03 Parametric Equations & Modeling

Find the component form of vector $\mathbf{u}$ directed from $(9, 12)$ to $(20, -2)$.

a. $\langle 12, -14 \rangle$

b. $\langle -11, 14 \rangle$

c. $\langle 11, -14 \rangle$

d. $\langle 11, 14 \rangle$

45 [Challenging] • Target: Exponential growth and decay basics • Cross-Lesson: 1.01 Polynomial End Behavior & Zeros

Does the point $(4, 0)$ satisfy the implicitly defined curve $x^2 + y^2 = 16$?

a. No, the point does not lie on the curve

b. The equation does not define a curve

c. Yes, the point lies on the curve

d. Only if r is negative

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